

IoT and AIML Workshop
Indian Institute of Information Technology, Sri City

IoT for Intelligent Transportation Systems and Autonomous Navigation



Bharatesh Chakravarthi, Ph.D.

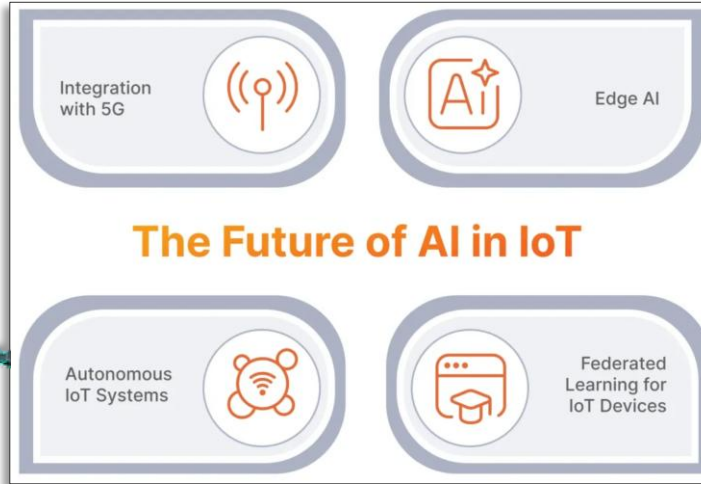
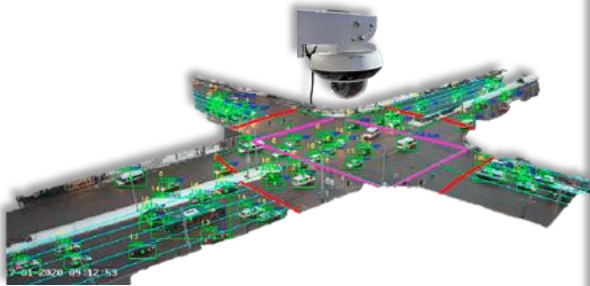
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Arizona State University, USA

01

***IoT and ITS:
A New Era of Technological Integration***

AI and IoT: A New Era of Technological Integration

Grasping the Internet of Things

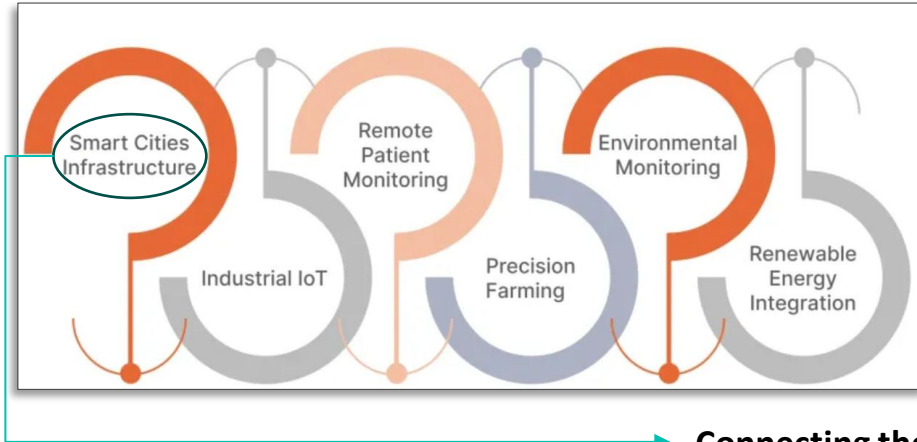


Advantages of AI in IoT



AI and IoT: A New Era of Technological Integration

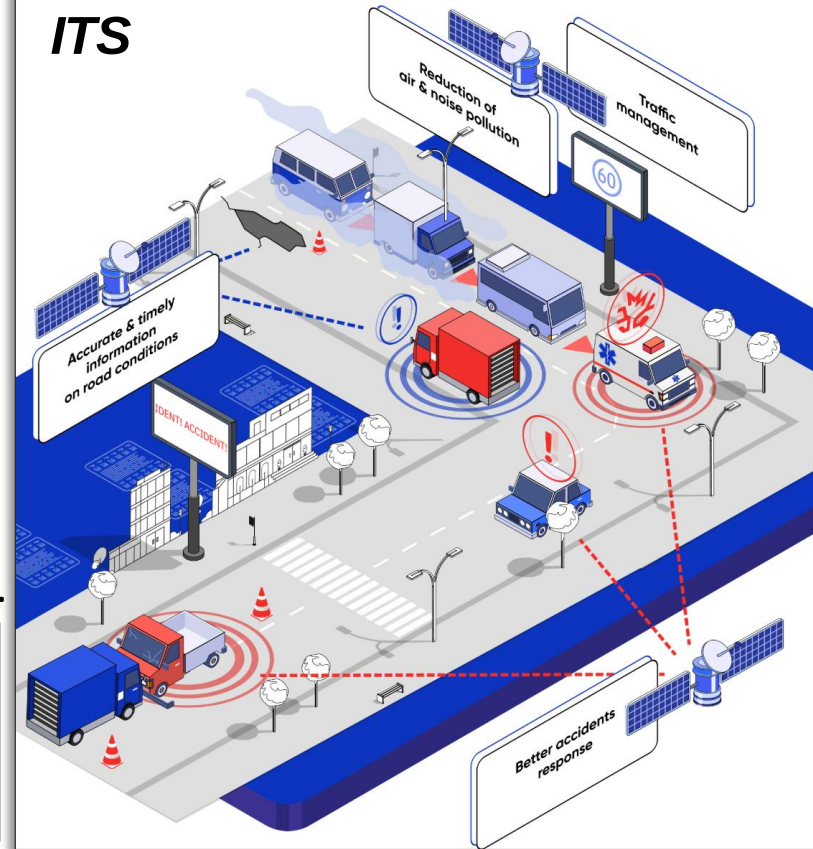
Prominent Applications of AI and IoT



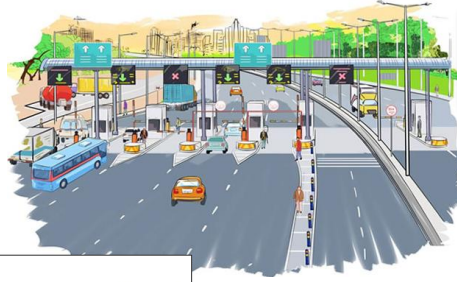
Connecting the dots...



ITS



AI and IoT: A New Era of Technological Integration



AI traffic system cuts travel time by a third at key junctions in Bengaluru

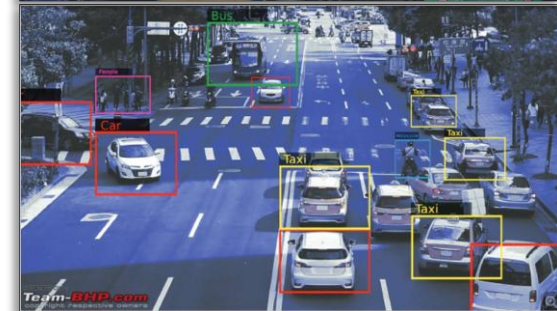
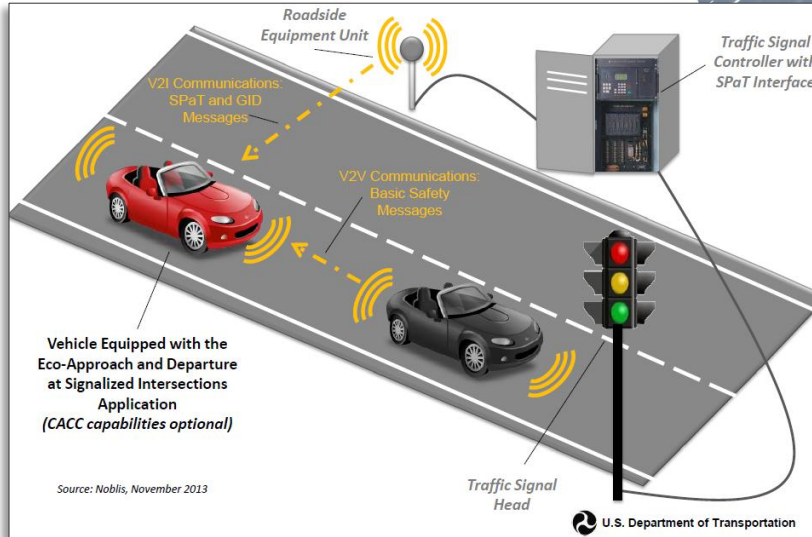
Cops plan to deploy AI in 250 out of 500 signals by March.

**Bengaluru police deploys AI-based traffic control system:
Here's how it works**

TIMESOFINDIA.COM / Oct 11, 2024, 11:13 IST

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ITS



AI and IoT: A New Era of Technological Integration



AI traffic system cuts travel time by a third at key junctions in Bengaluru

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TIMESOFINDIA.COM / Oct 11, 2024, 11:13 IST

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ITS



The Vehicle Actuated Control (VAC) mode is a feature of the AI-based traffic signal system in Bengaluru, India that adjusts traffic light timings based on the number of vehicles on the road:

How it works

Cameras and sensors at key junctions monitor the number of vehicles approaching from different directions. The system then adjusts the signal timings to improve traffic management.

When it's used

The system operates automatically 90% of the time, switching to VAC mode during peak hours. Manual mode is only used 5% of the time for emergency vehicles or VIPs.

Benefits

The system reduces the need for traffic police intervention and ensures that emergency vehicles can pass quickly.

Where it's used

The VAC mode is currently operational at junctions like KR Road, Rose Garden Road, and Hudson Circle.

AI and IoT: A New Era of Technological Integration



ITS



Automotive Training Centre

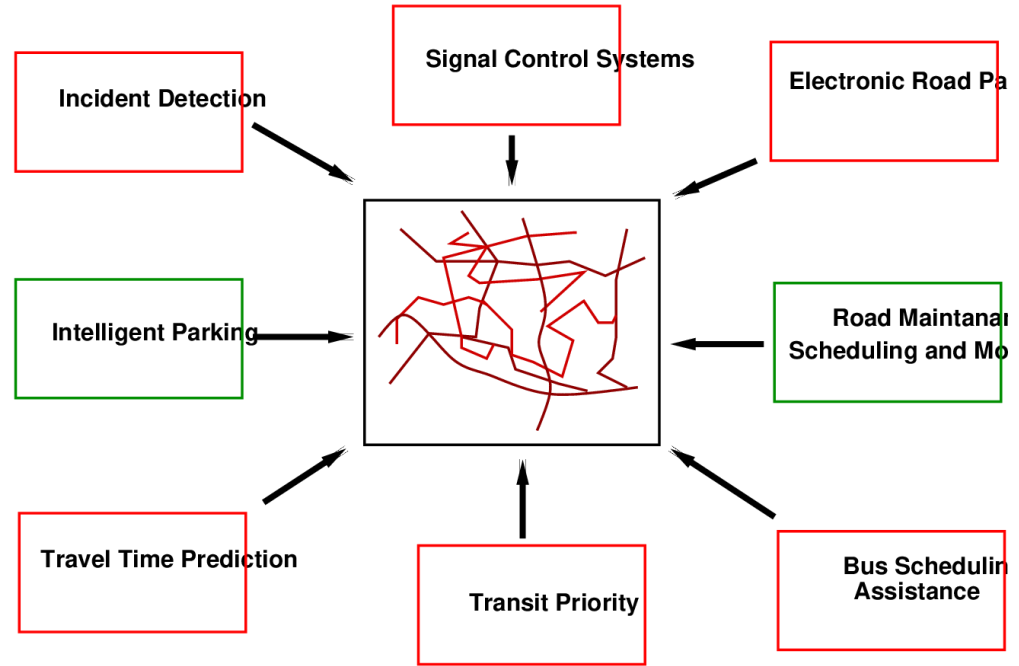
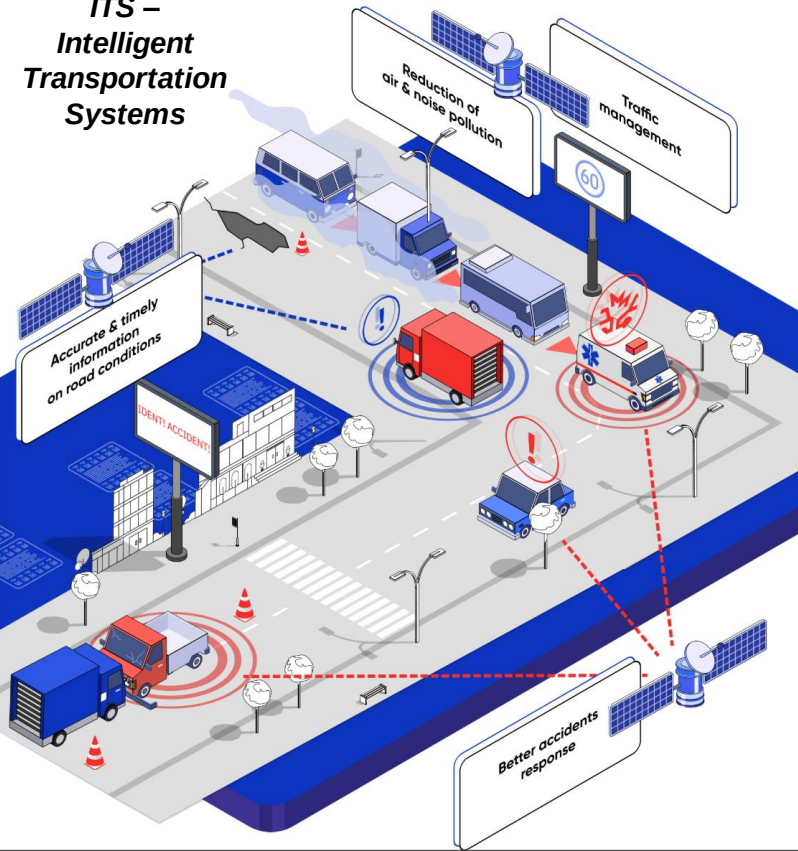
<https://www.autotrainingcentre.com> › blog › v2v-future... ⋮

V2V and the Future of "Talking Cars"

Talking cars are not purely theoretical or many years away from production. General Motors is already preparing to launch a full line of cars that incorporate ...

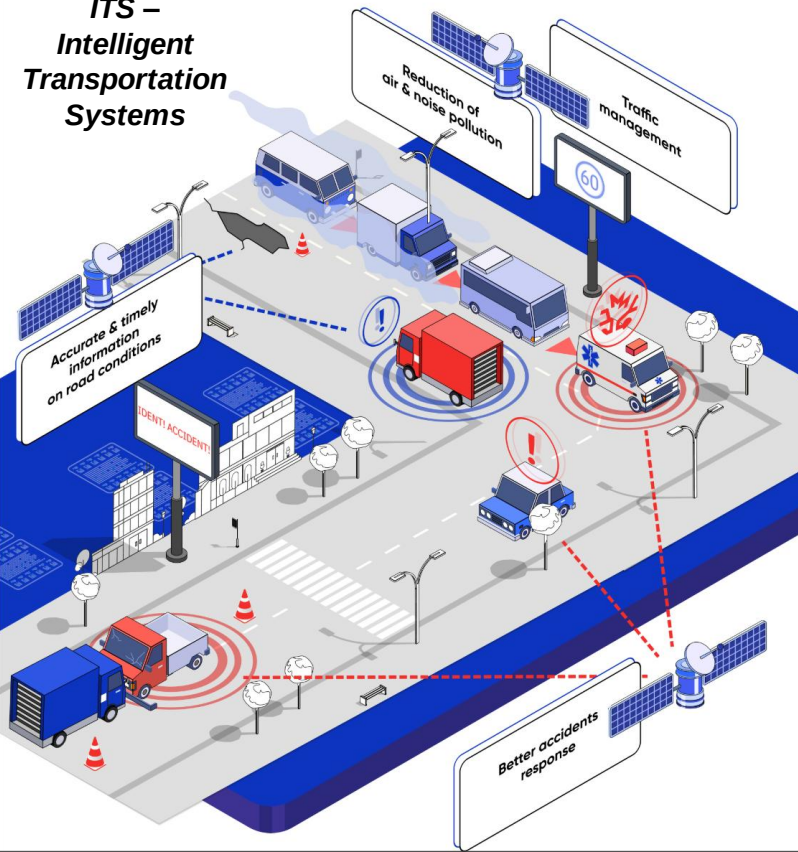
AI and IoT: A New Era of Technological Integration

ITS – Intelligent Transportation Systems



AI and IoT: A New Era of Technological Integration

ITS – Intelligent Transportation Systems

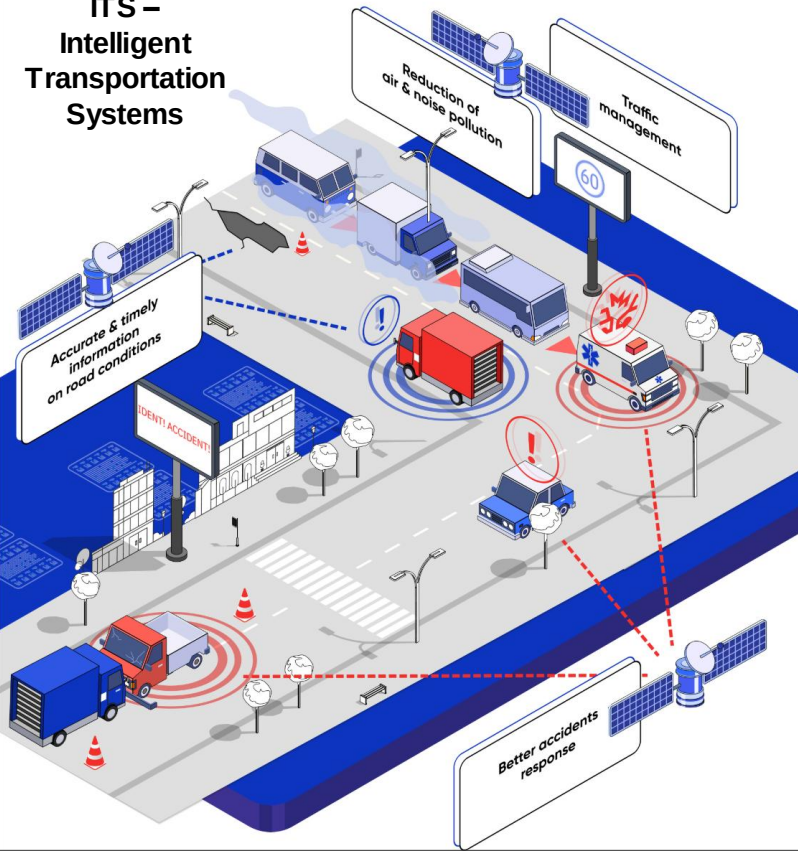


Autonomous Vehicles



AI and IoT: A New Era of Technological Integration

ITS – Intelligent Transportation Systems

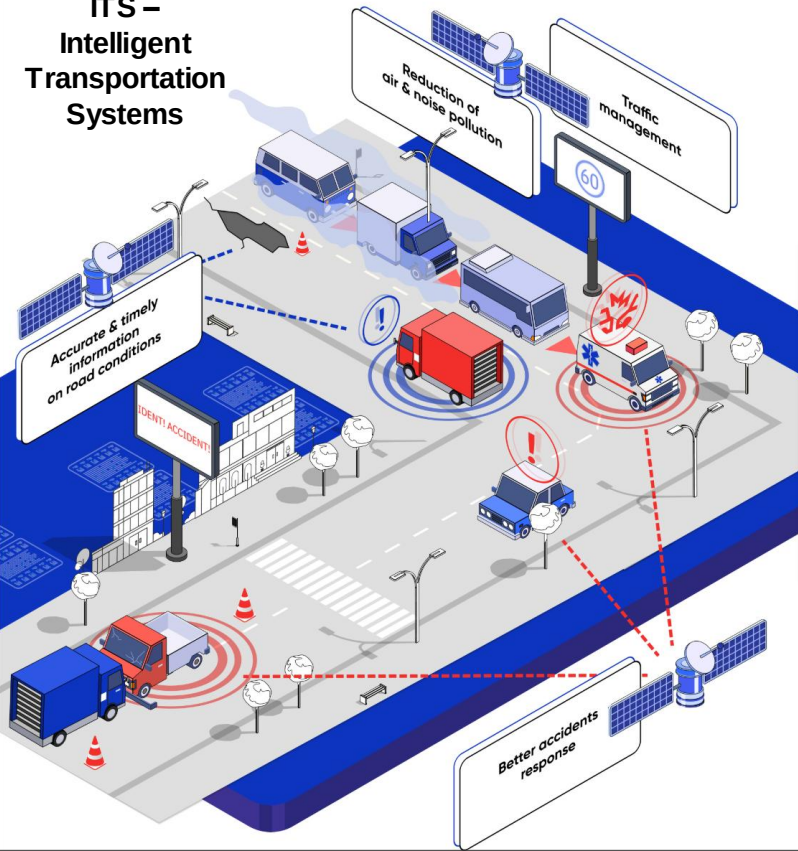


Autonomous Route Management



AI and IoT: A New Era of Technological Integration

ITS – Intelligent Transportation Systems

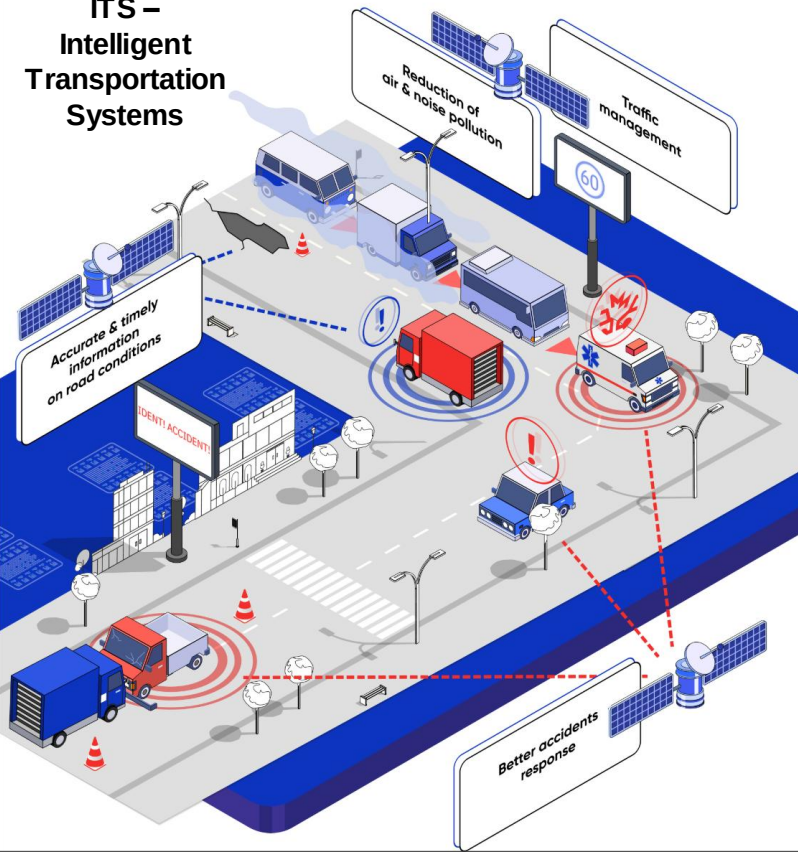


Traffic Incident Management



AI and IoT: A New Era of Technological Integration

ITS – Intelligent Transportation Systems

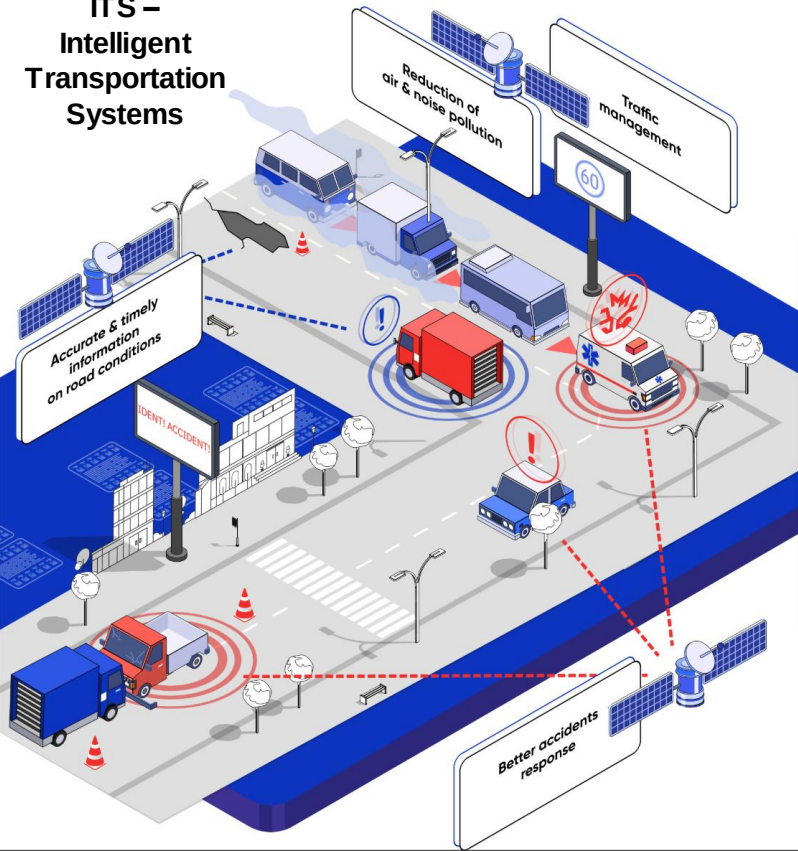


Automatic Traffic Intersection Monitoring

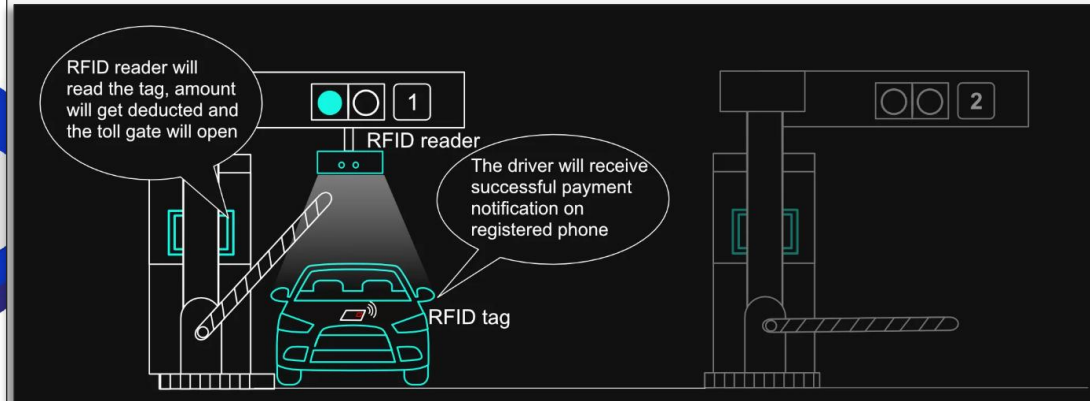


AI and IoT: A New Era of Technological Integration

ITS – Intelligent Transportation Systems



Automated Toll Management

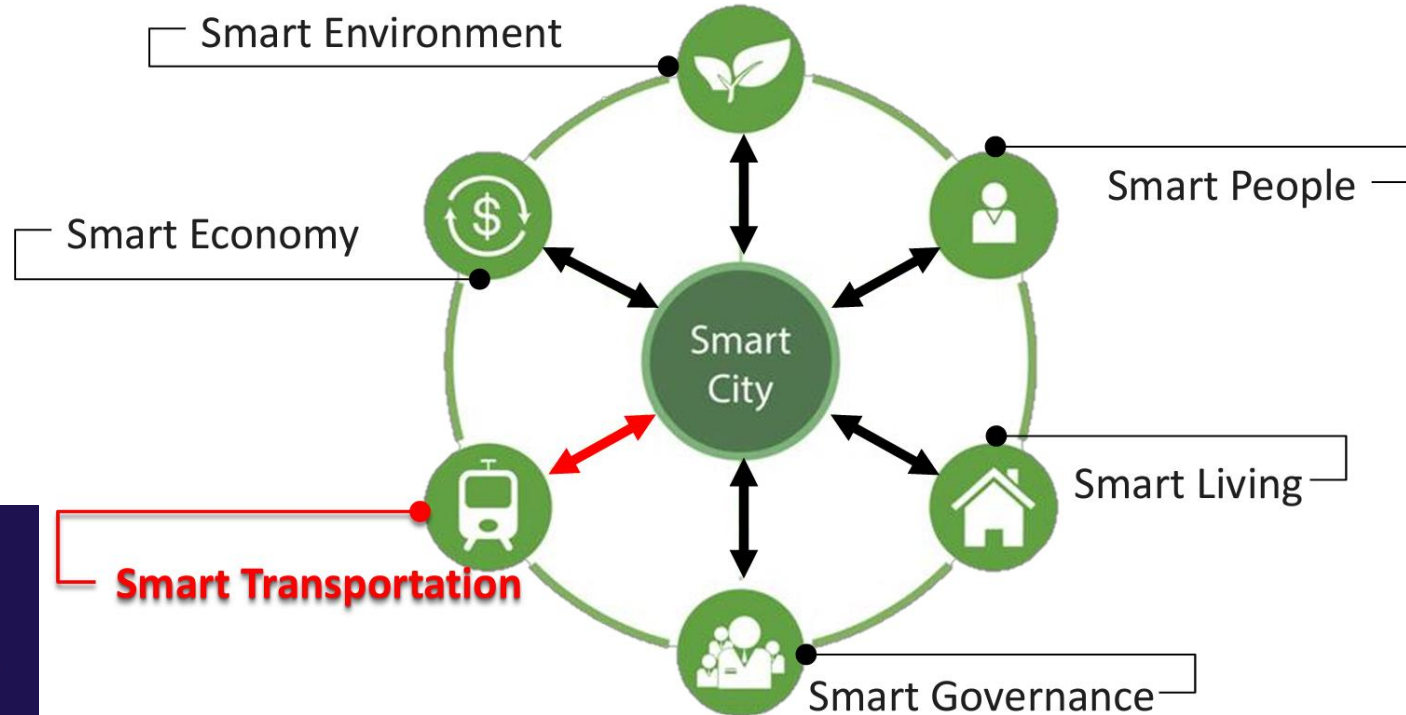


02

IoT and ITS-enabled Smart Cities

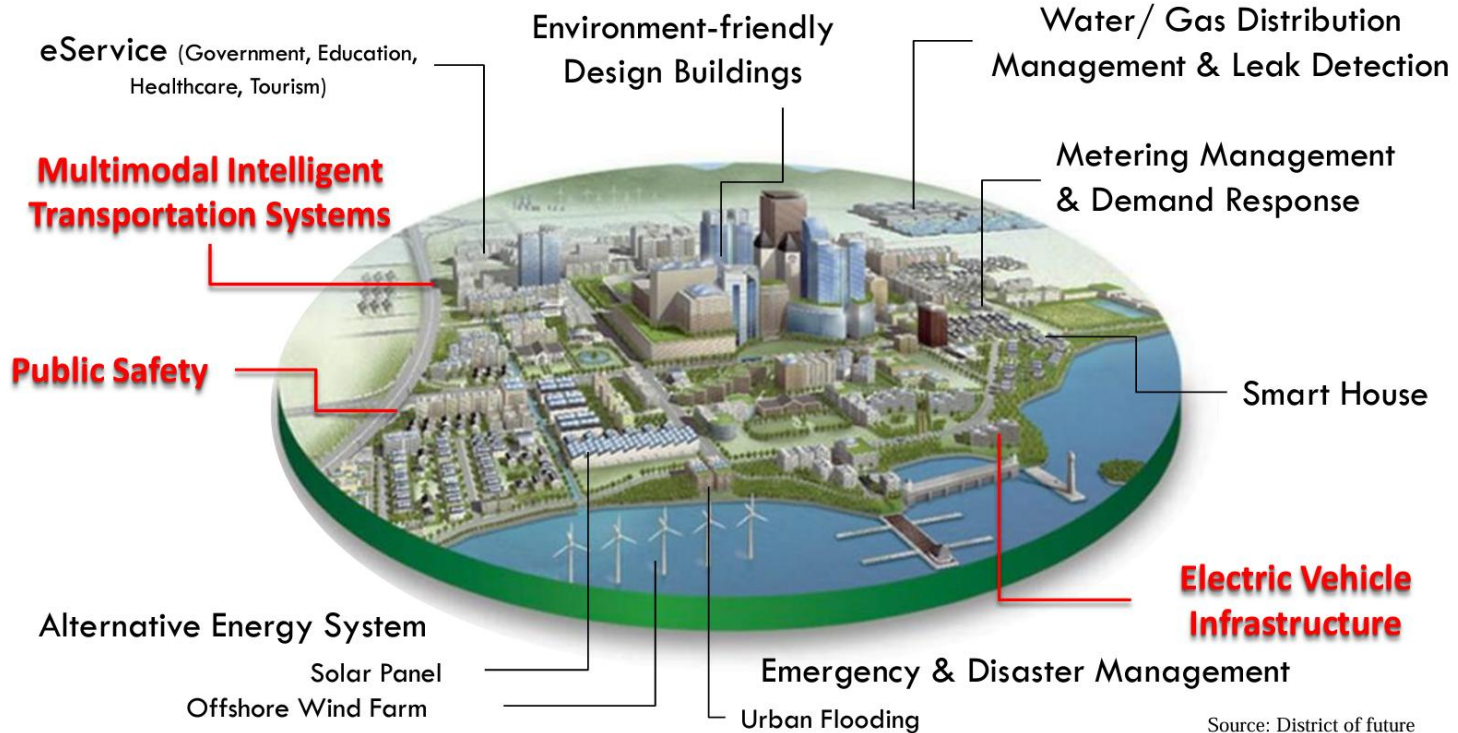
IoT and ITS enabled Smart Cities

“Smart city” – the use of technologies to provide intelligent response to the needs of the city. *(Smart City and the Applications, 2011)*



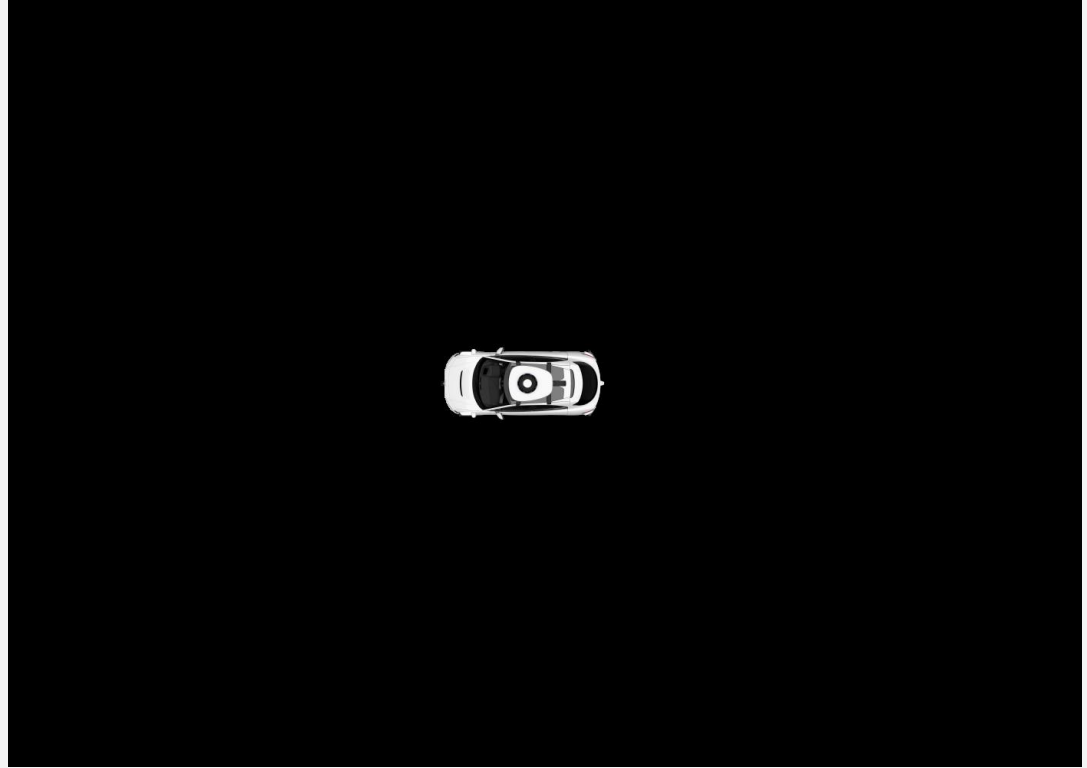
IoT and ITS enabled Smart Cities

What are the needs for developing a smart city in the future?



Source: District of future
<http://www.districtoffuture.eu/>

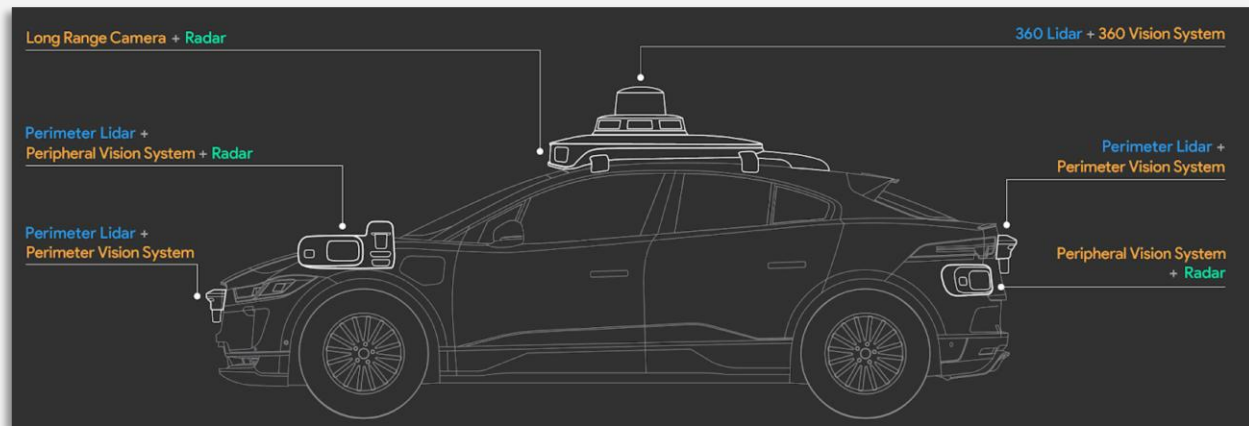
IoT and ITS enabled Smart Cities



IoT and ITS enabled Smart Cities

August 19, 2024

Meet the 6th-generation Waymo Driver: Optimized for costs, designed to handle more weather, and coming to riders faster than before

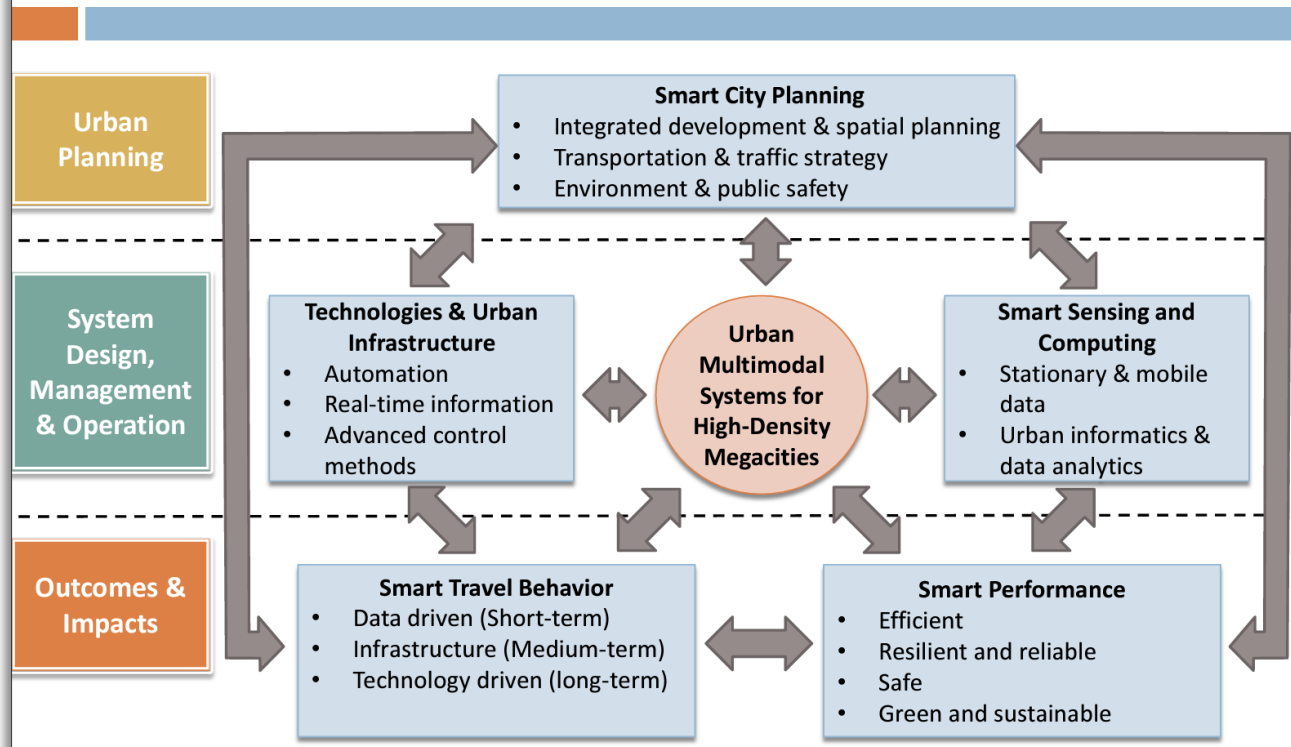


- **Cameras:** The sixth generation has 13 cameras instead of 29.
- **Lidar:** The sixth generation has 4 lidar sensors instead of 5.
- **Radar:** The sixth generation has 6 radar devices. [Link](#)

Our 6th-gen system builds on the unparalleled capabilities of our current, 5th-generation system, which has been instrumental to helping Waymo scale our service to some of the densest cities in the United States and [improving road safety](#) where we operate. With 13 cameras, 4 lidar, 6 radar, and an array of external audio receivers (EARs), our new sensor suite is optimized for greater performance at a significantly reduced cost, without compromising safety. It provides the Waymo Driver with overlapping fields of view, all around the vehicle, up to 500 meters away, day and night, and in a range of weather conditions.

IoT and ITS enabled Smart Cities

Smart Transportation is About



IoT and ITS enabled Smart Cities



Adaptive signal control



Bicycle sharing system

- Is high-density development a solution to rapid urbanization?
- What are the enabling technologies and urban infrastructure to enhance sustainability, accessibility, mobility, and wellbeing?



Automobile sharing



Walking/Cycling facilities

IoT and ITS enabled Smart Cities



Autonomous self-driving vehicle

➤ How would autonomous vehicles modify fundamental traffic flow properties, and impact infrastructure design and urban form?

➤ How would electric vehicles interface with the smart grid in terms of energy distribution and storage?



Electric vehicle charging infrastructure



Lo Wu	Platform 4	14:53
E.TST to Tuen Mun	Platform 3	14:54
Lo Wu	Platform 1	14:59
E.TST to Tuen Mun	Platform 2	15:01
Lok Ma Chau	Platform 4	15:03
Lo Wu	Platform 1	15:06

Real-time travel information



Speed Map Panel

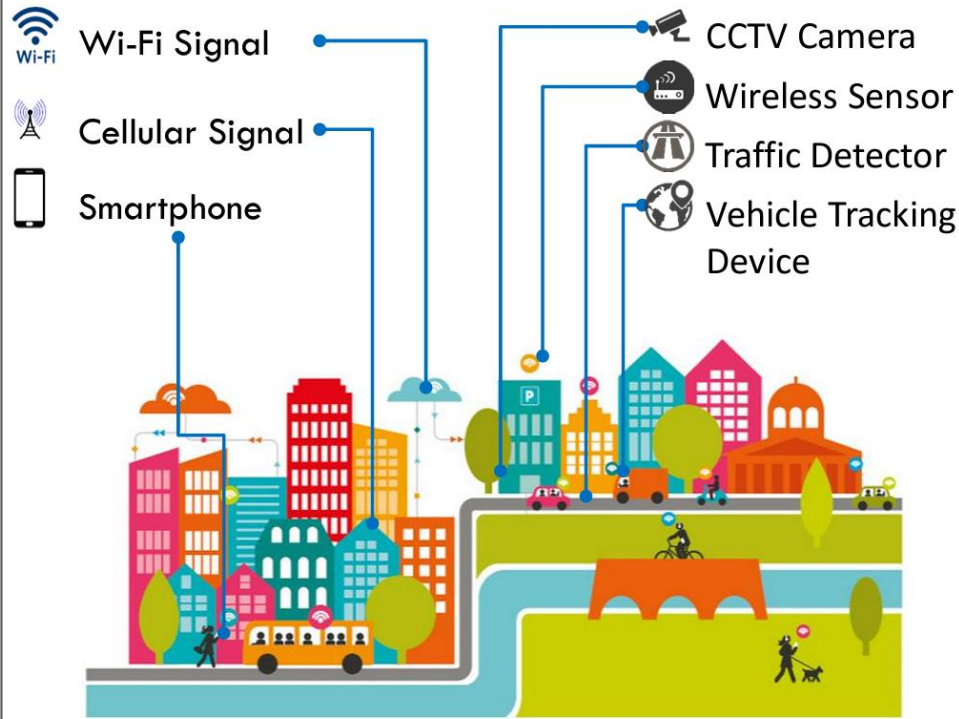


Journey Time Indicator

03

The IoT Aspect of ITS

Smart Sensing



Source: CloudT Project
<http://clout-project.eu/>

- What are the sensing strategies for collecting stationary and mobile sources of multi-modal traffic data and how are these data integrated and interpreted?
- What are the computing strategies for centralized and distributed data transmission, processing, interfacing, analysis, sharing, dissemination, and storage, in the context of big data arena?

Smart Travel Behavior

Data driven



Short-term

Medium-term

Long-term

Technology driven

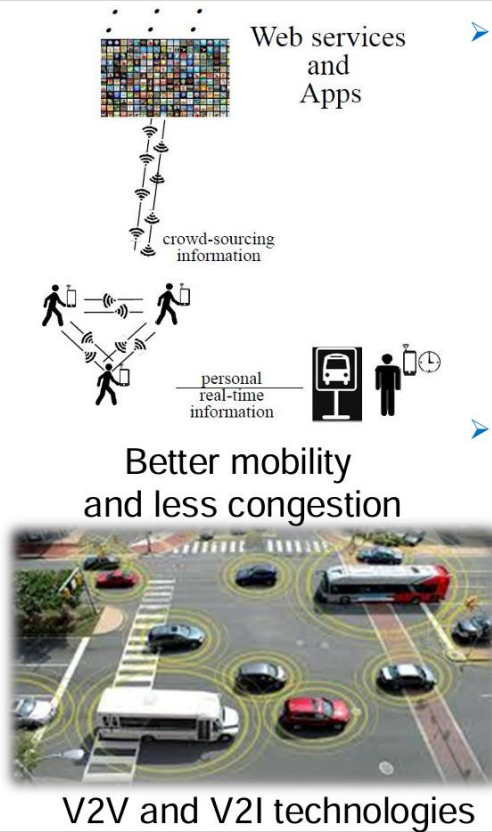


Infrastructure driven



- How would accurate, reliable and timely multi-modal traffic information affect travelers' decision making processes?
- How would future technologies, such as autonomous self-driving vehicles, electric vehicles, multi-modal traffic information, massive and robust traffic control affect activity and mobility patterns?

Smart Performance



➤ How to develop a cost-effective but highly resilient multi-modal transportation system in response to increasingly frequent and serious natural and manmade disruptions?

➤ How would the above smart developments help to maintain safe, healthy, rapid, reliable, comfortable, convenient, affordable, equitable, and environmentally compatible mobility of mankind?

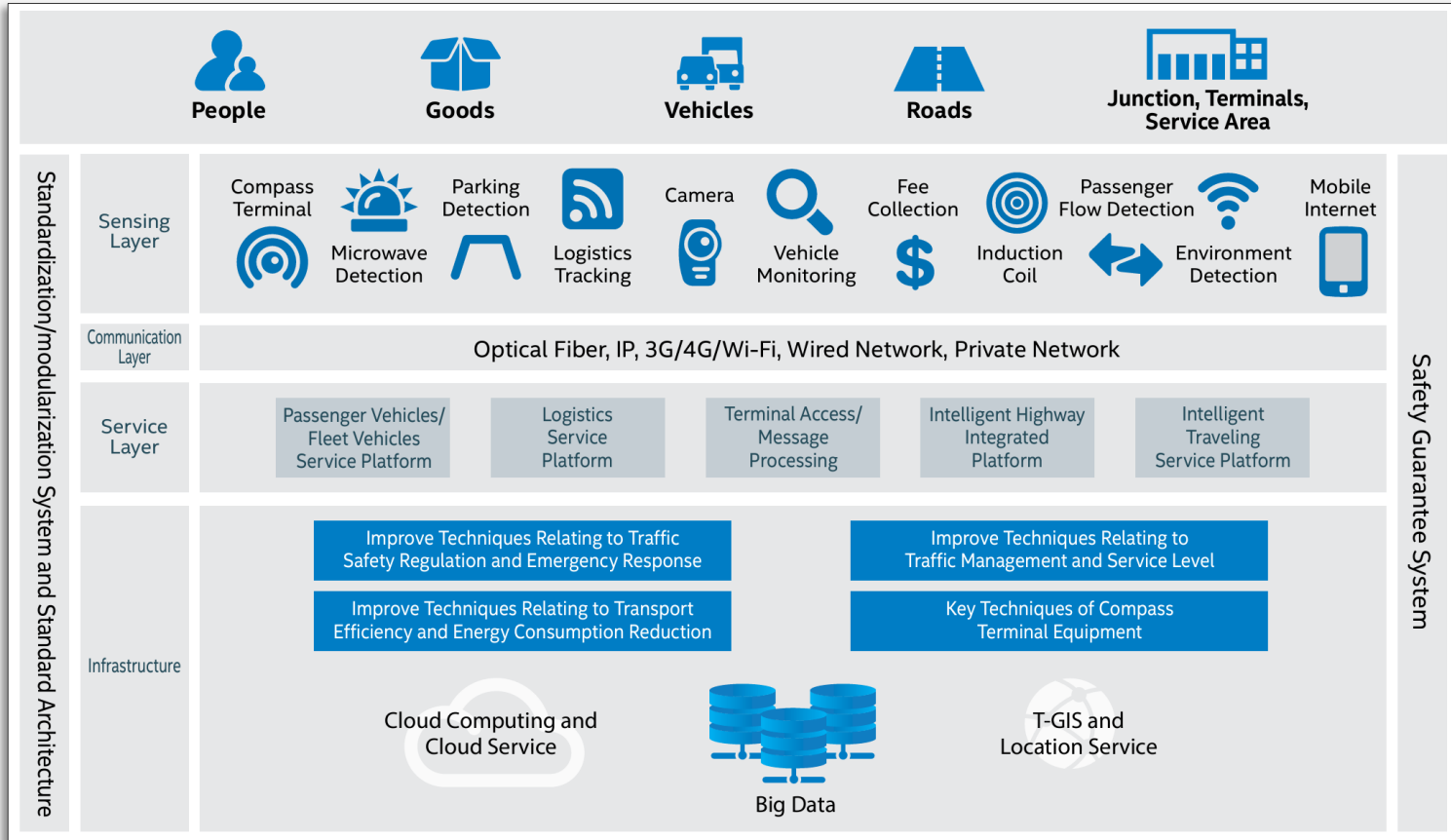


Uncertainties and disruptive conditions

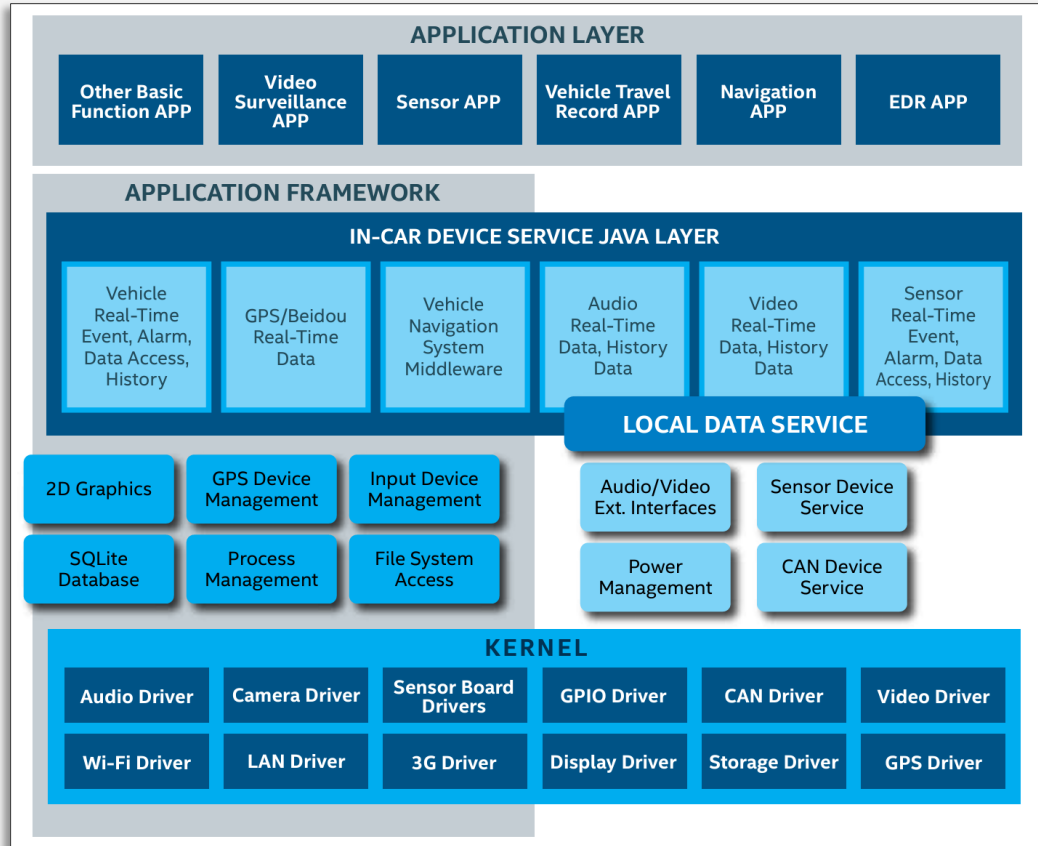


Smart vehicles

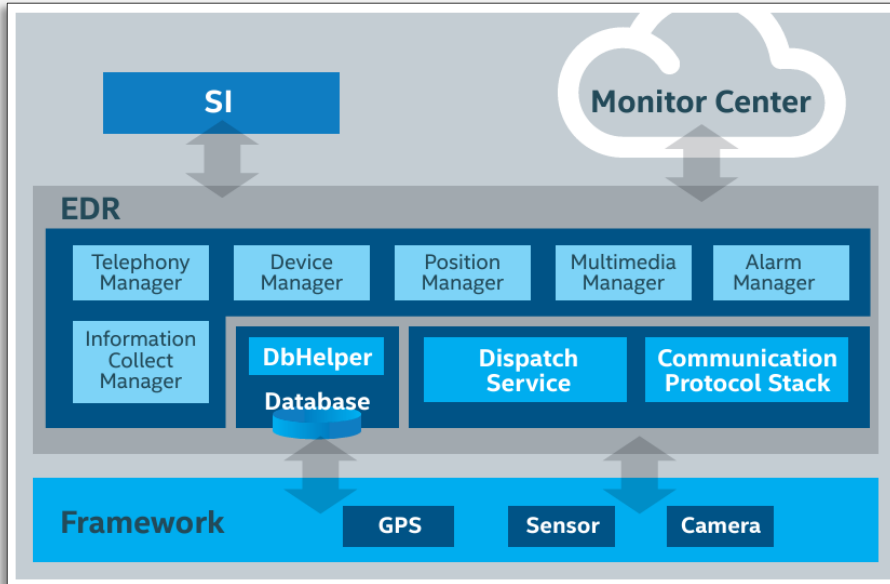
The Big Picture – IoT and ITS



Building an ITS system with IoT



Building an ITS system with IoT



Video Preprocessing
Video Encoding, Decoding,
Transcoding Video Post-Processing

Video Analytics
Application

Intel® Media SDK

Intel® SDK for OpenCL*
Applications

Hardware Library for Fixed Function/EU Acceleration

Graphics Driver

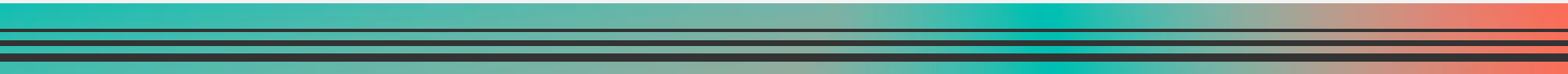
3rd/4th Gen Intel® Core™ Processor
Intel® Xeon® Processor E3-1285/1285L
Intel® Atom™ Processor E3800 Series

Future
Intel® Architectures

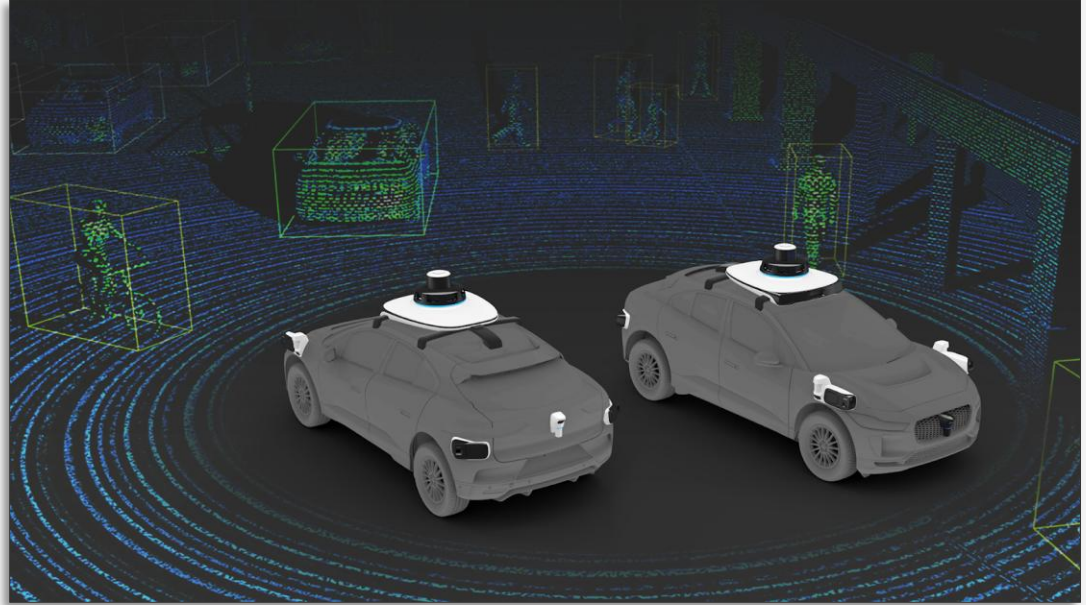
04

Case Studies

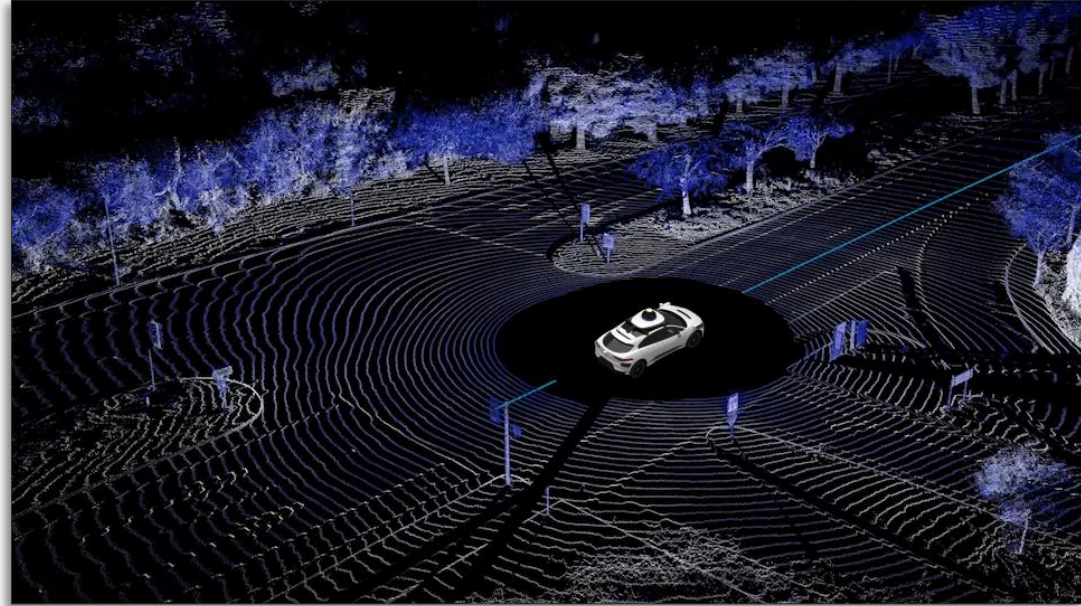
Building an ITS system with IoT

- 1. Define the Scope and Objectives*
 - 2. Select IoT Devices and Sensors*
 - 3. Data Acquisition and Processing*
 - 4. Develop Intelligent Algorithms and Analytics*
 - 5. Build a Robust IT Infrastructure*
 - 6. Deploy and Integrate with Existing Systems*
 - 7. Monitor and Optimize*
- 

Building an ITS system with IoT



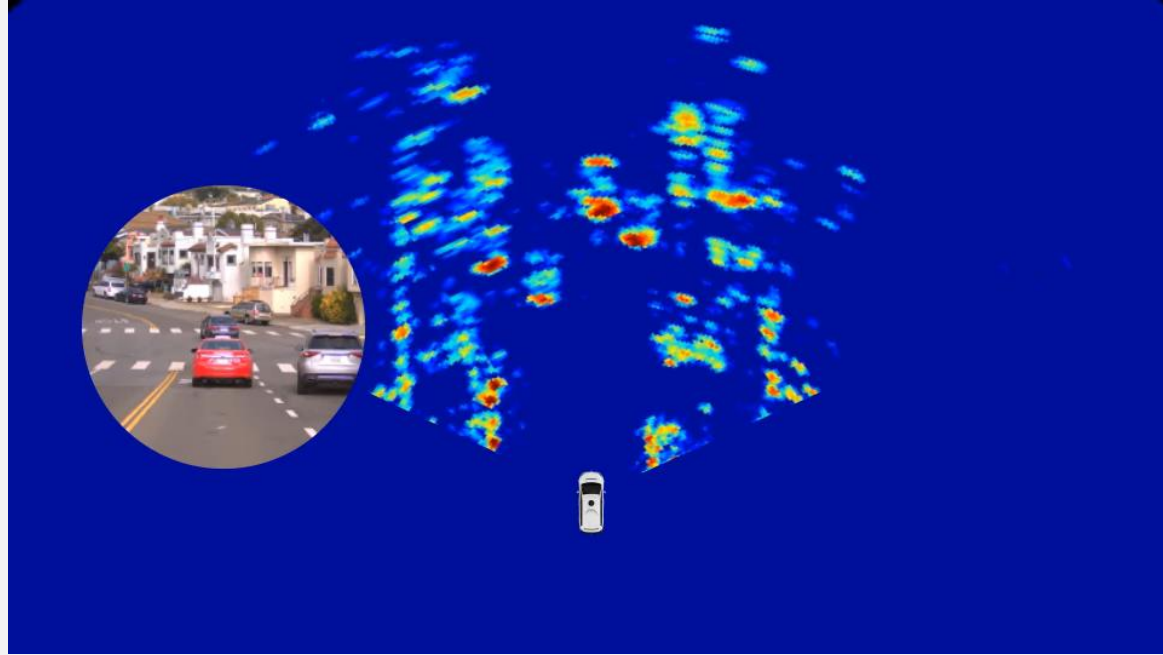
Building an ITS system with IoT



Building an ITS system with IoT



Building an ITS system with IoT



Building an ITS system with IoT – Sensing GAP

Multi-Sensor Modality

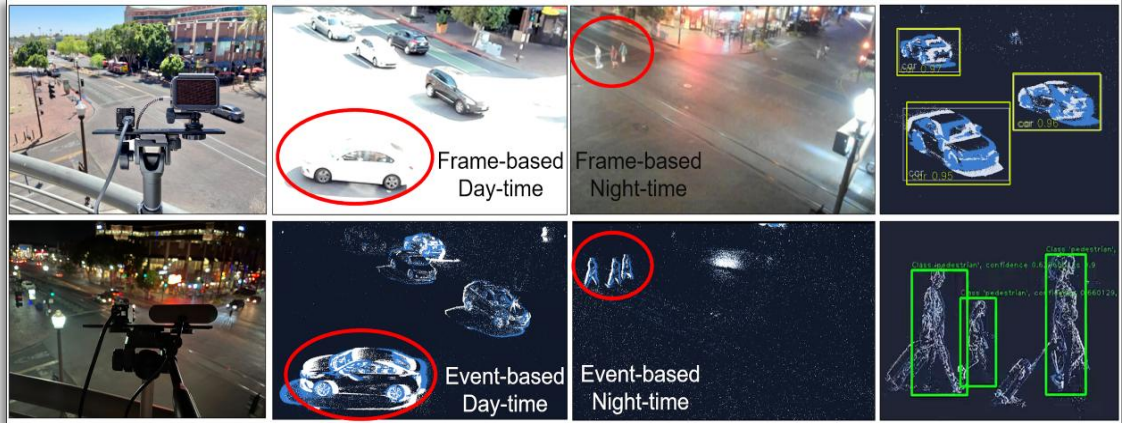
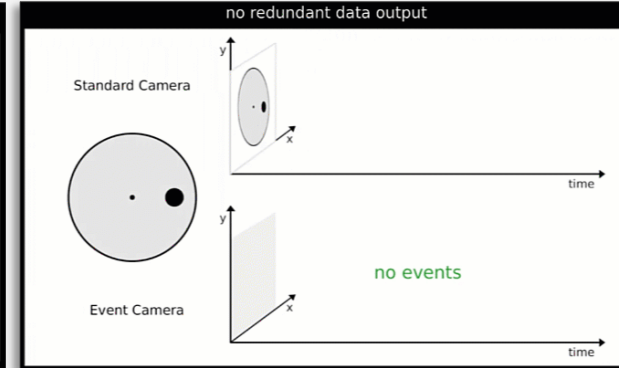
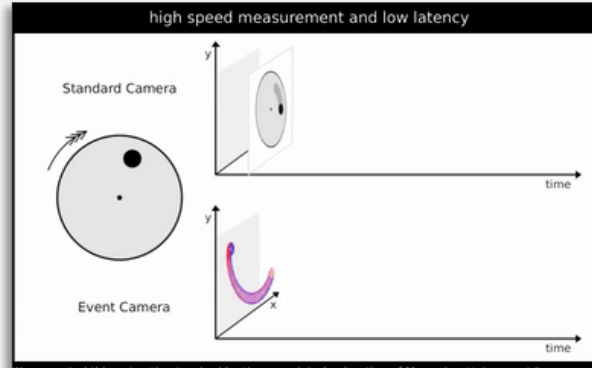
- Neuromorphic Vision, Monocular Vision, and Synthetic Traffic Data
- Fusion for detection, tracking and prediction

Improved Performance

- Diverse Traffic Participants,
- Improved Performance in Challenging Conditions (Lighting & Weather)

Vision-based Alert System

- Vehicle Trajectory Prediction
- VRU's Alert & Safety System for Signalized Intersections
- Dynamic alert signs / Connected vehicles & Argos Vision Infrastructure Integration



Building an ITS system with IoT – Sensing GAP

Multi-Sensor Modality

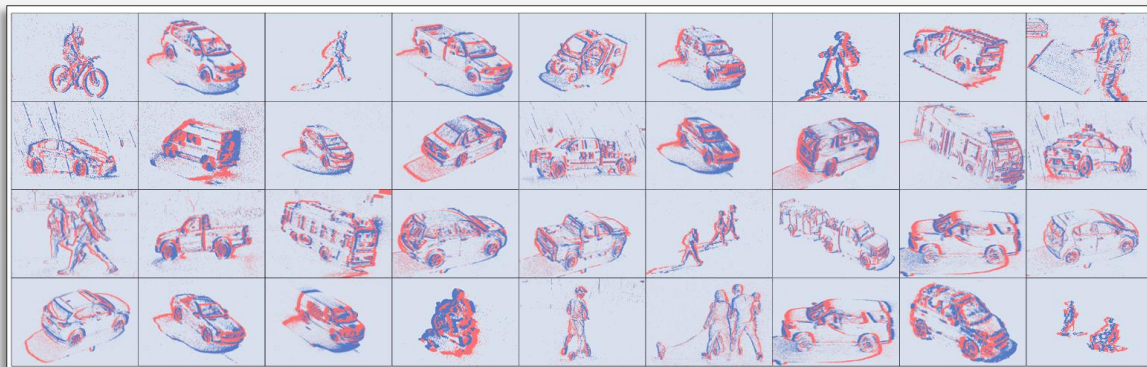
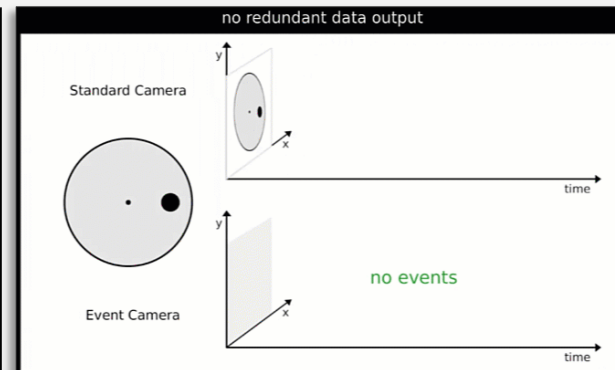
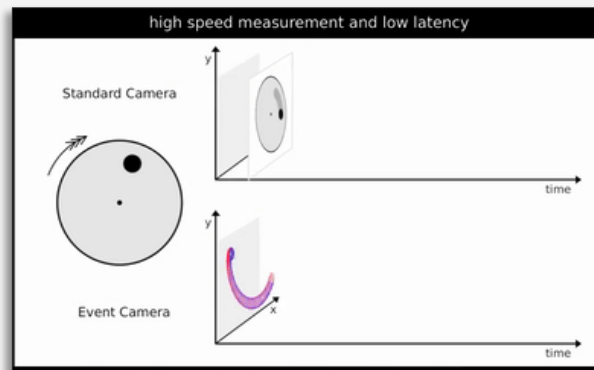
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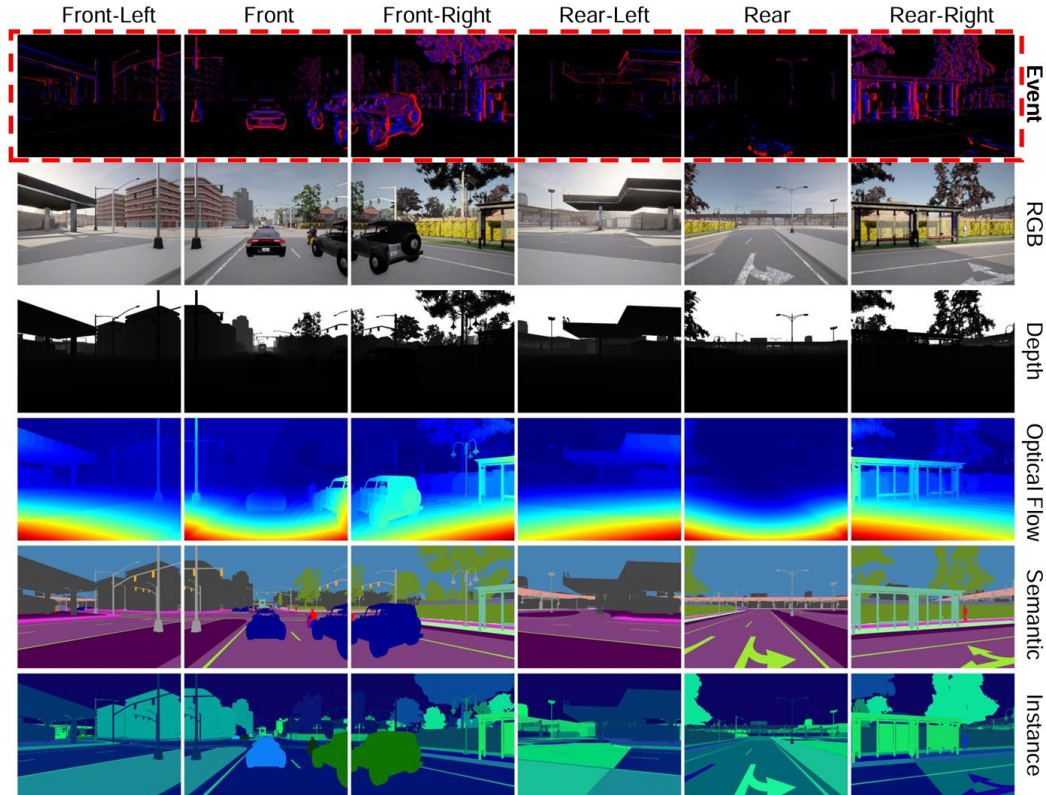


Building an ITS system with IoT – Sensing GAP

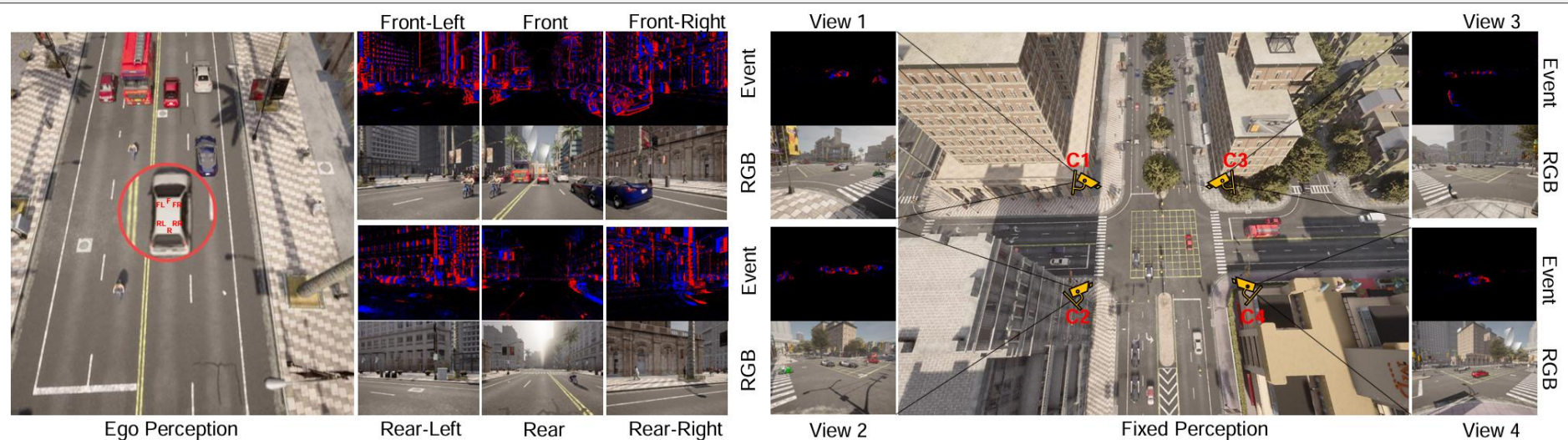
Sensor Types = {Event,
RGB, Depth, Optical
Flow, Semantic,
Instance, GNSS, IMU}



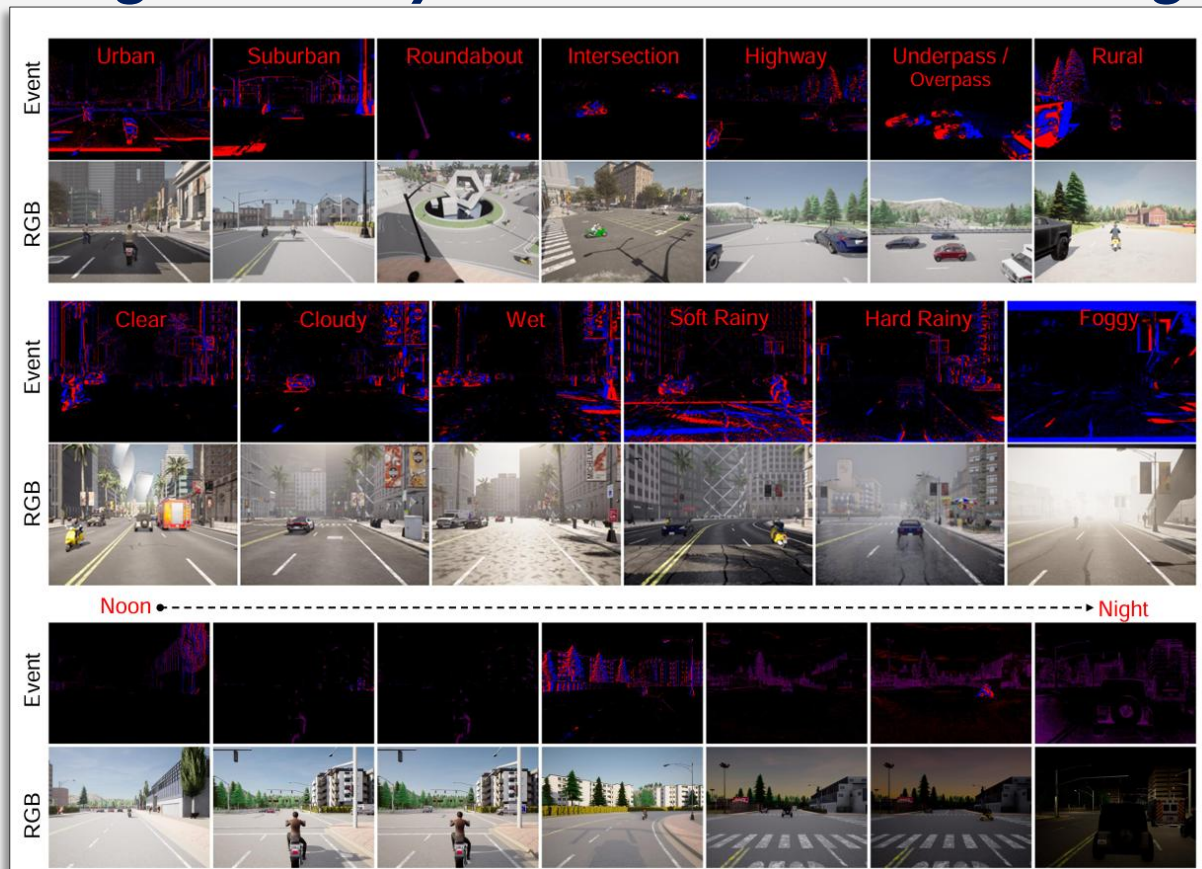
Camera Position =
{Front(F), Front-
right(FR), Front-Left(FL),
Rear(R), Rear-right(RR),
Rear-Left(RL)}



Building an ITS system with IoT – Sensing GAP



Building an ITS system with IoT – Sensing GAP





Thank You All
